

ETP formalization benchmark · synthetic

laws to 10 ops/equation (no-think)

Equational Theories Project implications rendered per each run's form, formalized back by each model over OpenRouter, and graded syntactically (checkform.py). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

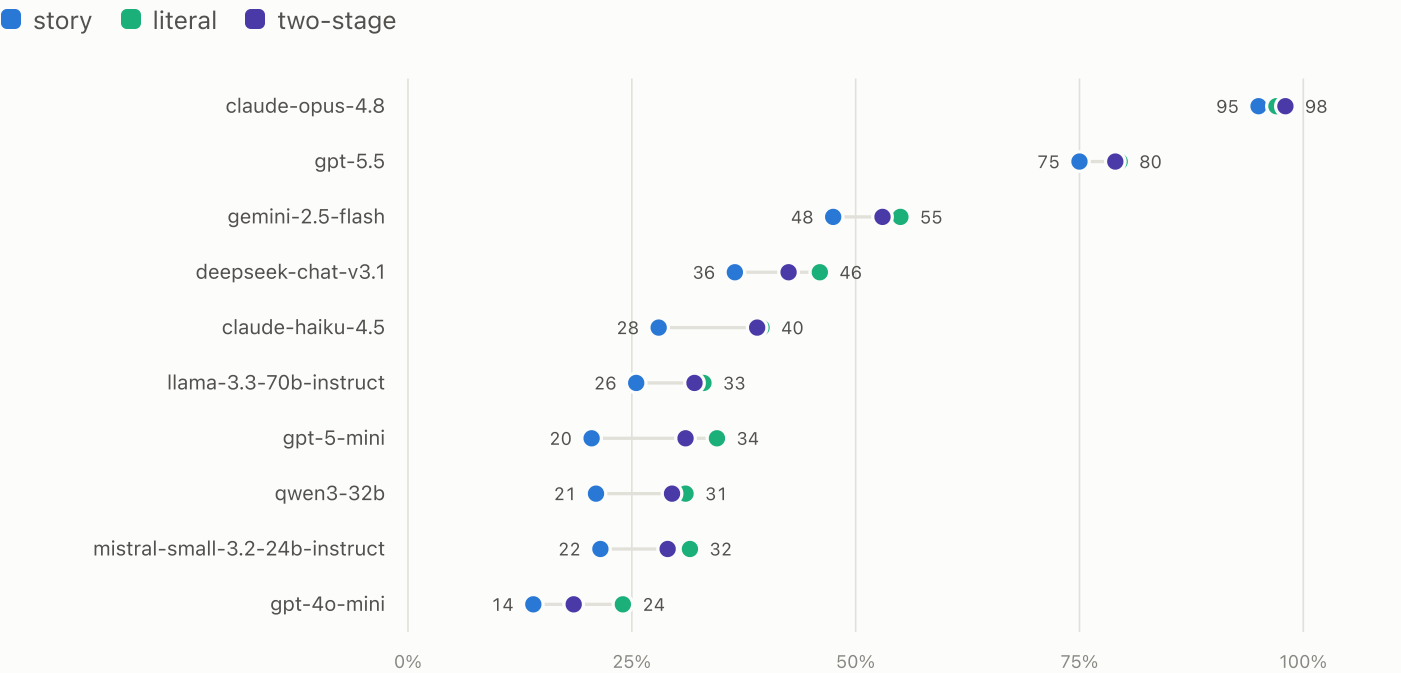
experiments/09-synthetic-complexity/runs/run-eqstrat20-s0-literal-think-off · experiments/09-synthetic-complexity/runs/run-eqstrat20-s0-story-think-off · experiments/09-synthetic-complexity/runs/run-eqstrat20-s0-two-stage-think-off

Runs	Models	Stories	Graded calls	Overall correct
3	10	200	6,000	44%

200 pairs, 20 per per-equation operation bin 1–10, both laws from the same bin · seed 0 · story vs literal vs two-stage · no-think

FIGURE 1 Correct rate by model and form — no-think · balanced 200

The same pairs rendered per arm; each row compares one model across the forms under the no-think regime. Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

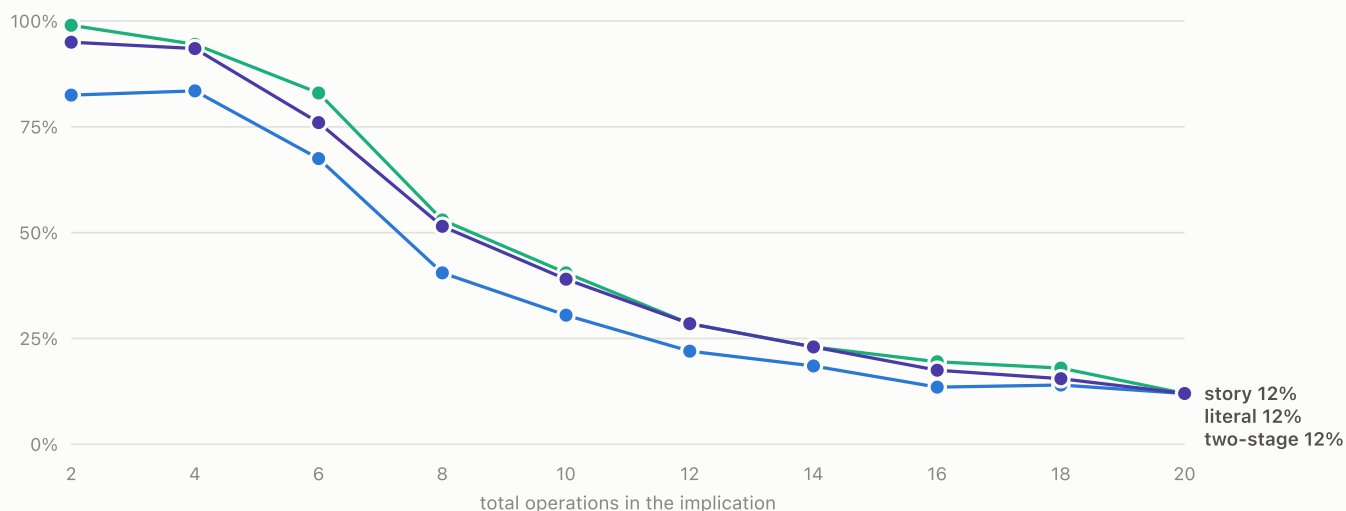


► Data: correct rate per model and form

FIGURE 2 Accuracy vs operation count — no-think · balanced 200

Correct rate pooled over all models, by the implication's total operation count (both equations combined); one line per form, no-think regime. Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

■ story ■ literal ■ two-stage

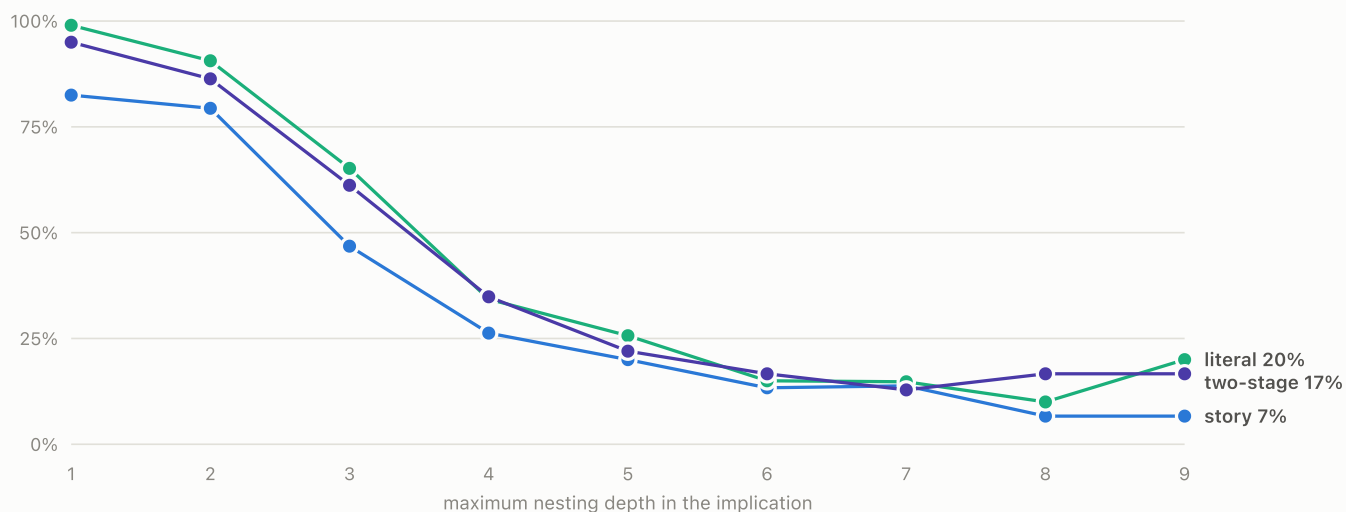


► Data: pooled correct rate per operation-count bin

FIGURE 3 Accuracy vs nesting depth — no-think · balanced 200

Correct rate pooled over all models, by the deepest operation nesting across the four equation sides; one line per form, no-think regime. Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

■ story ■ literal ■ two-stage



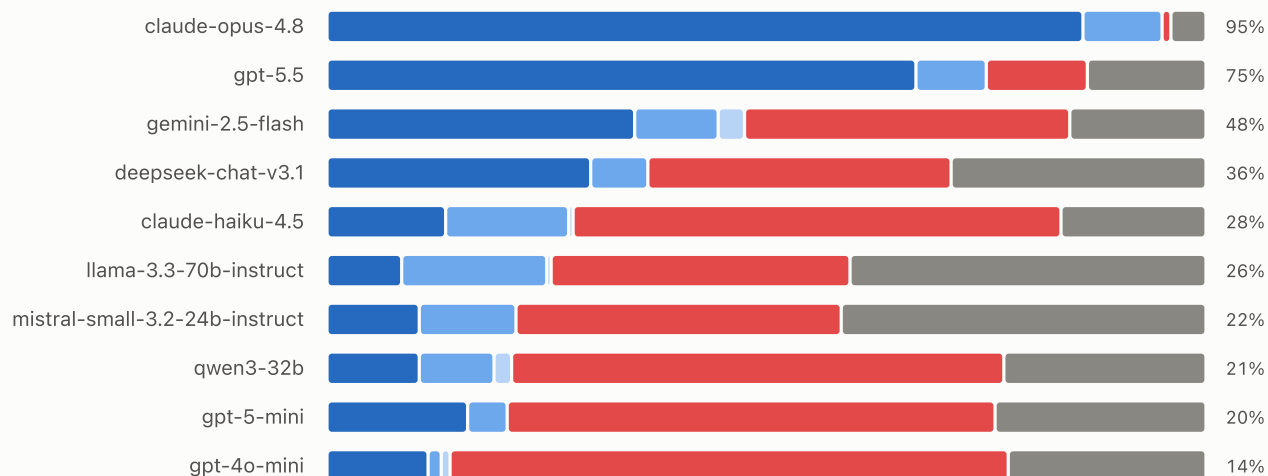
► Data: pooled correct rate per depth bin

FIGURE 4 **Verdict composition — story · no-think · balanced 200**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



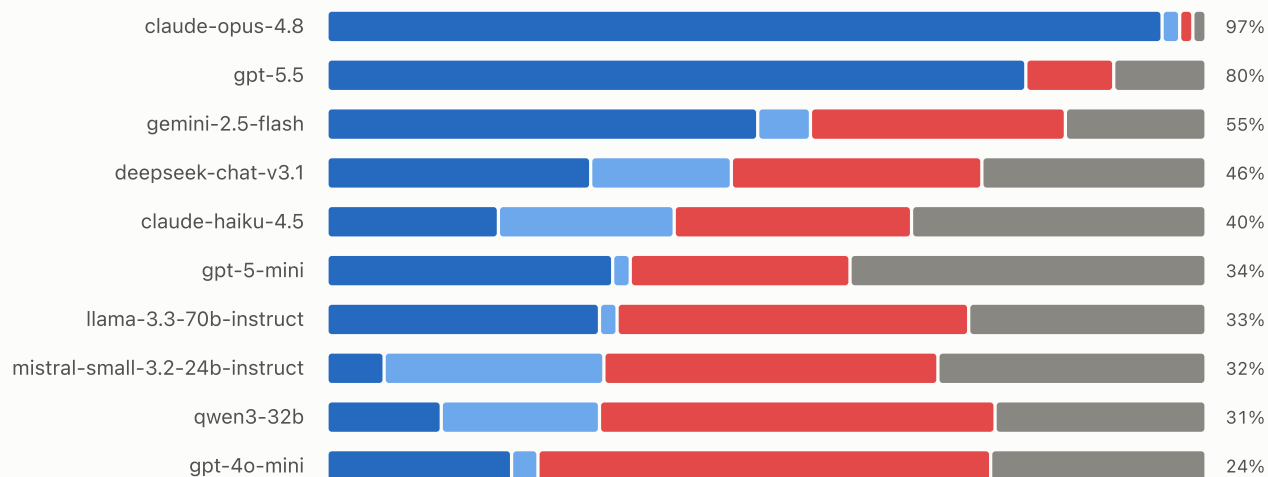
► Data: verdict counts per model

FIGURE 5 **Verdict composition — literal · no-think · balanced 200**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

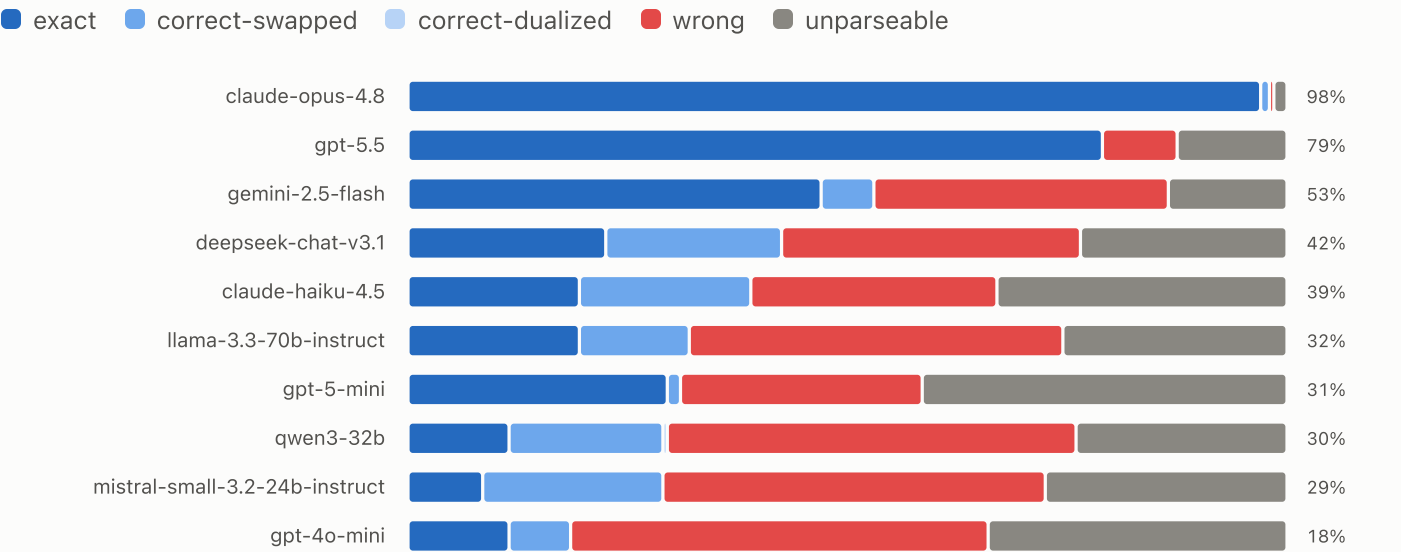
■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 6 **Verdict composition — two-stage · no-think · balanced 200**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).
Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.



► Data: verdict counts per model